Computer Science & IT

COMPUTER NETWORKS (CN)

Flow Control Methods

Lecture No. 1



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Flow Control

Topics to be Covered



- Topic
- Topic
- Topic

Flow Control

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Inspiring Stories : Pidamarthi Anil Kumar

Pidamarthi Anil Kumar

Background: Son of daily-wage laborers in a remote Telangana village; his parents worked long hours with limited means.

Education: B.Tech at IIT Kharagpur, followed by advanced technical work in EV innovation.

Career Achievements: Co-co-founded Eloop Charging Solutions in Bengaluru. Developed the Loop Charging Module (LCM), a device that harnesses ambient airflow to generate electricity, enabling EVs to charge while driving.

Impact: His invention can increase EV mileage by up to 70%; a breakthrough in clean mobility born from humble roots and technical grit.



Inspiring Stories : Phoolan Devi

Phoolan Devi – The Bandit Queen Who Fought Back

Background: Born into a lower-caste family in rural India. Married off at 11. Abused, tortured, cast aside.

Struggles: Kidnapped by dacoits. Later led a bandit gang herself — targeted upper-caste oppressors.

Achievements: Surrendered, served time, and then became a Member of Parliament — still carrying scars, but holding a seat of power.

Impact: She turned trauma into rebellion. If she could fight a caste empire with rage and resolve, you can fight your own obstacles with code and courage.



Inspiring Stories : Shailesh Gandhi

Shailesh Gandhi Background: Modest upbringing, middle-class Maharashtra; began at IIT with documentary fame (I Am 20) but no industrial influence.

Education: B.Tech from IIT Mumbai, followed by legal and public policy credentials (including Master-level civic leadership training).

Career Achievements: Left a thriving business at age 56 to lead the Right to Information (RTI) campaign in India; served as Central Information Commissioner, adjudicating over 20,000 cases in under four years.

Impact: Played a pivotal role in enacting India's RTI Act (2005), empowering citizens with transparency tools. He sacrificed personal success for systemic reform.



Inspiring Stories : Krysten Anderson

Krysten Anderson – The Girl Who Took Over Grave Digger

Background: Daughter of a legendary monster truck driver. Grew up in the pits — not on the track.

Struggles: No one thought a woman could drive Grave Digger, the most iconic monster truck. Faced criticism, sexism, and physical crashes.

Achievements: Now holds multiple records in Monster Jam history — including longest jump by a woman and first female to drive Digger full time.

Impact: She didn't break barriers. She drove over them.

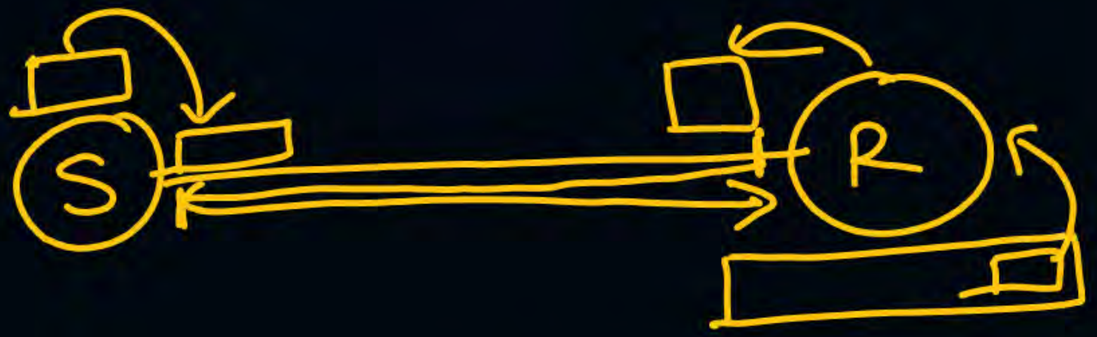
If she could rewrite the rules in a man's sport, you can rewrite your future when people around you are doubting you.

$$T_t = \frac{L}{B}$$

$$T_p = \frac{d}{V}$$



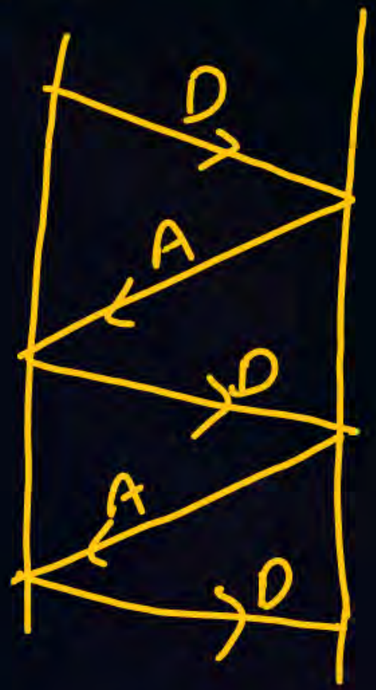


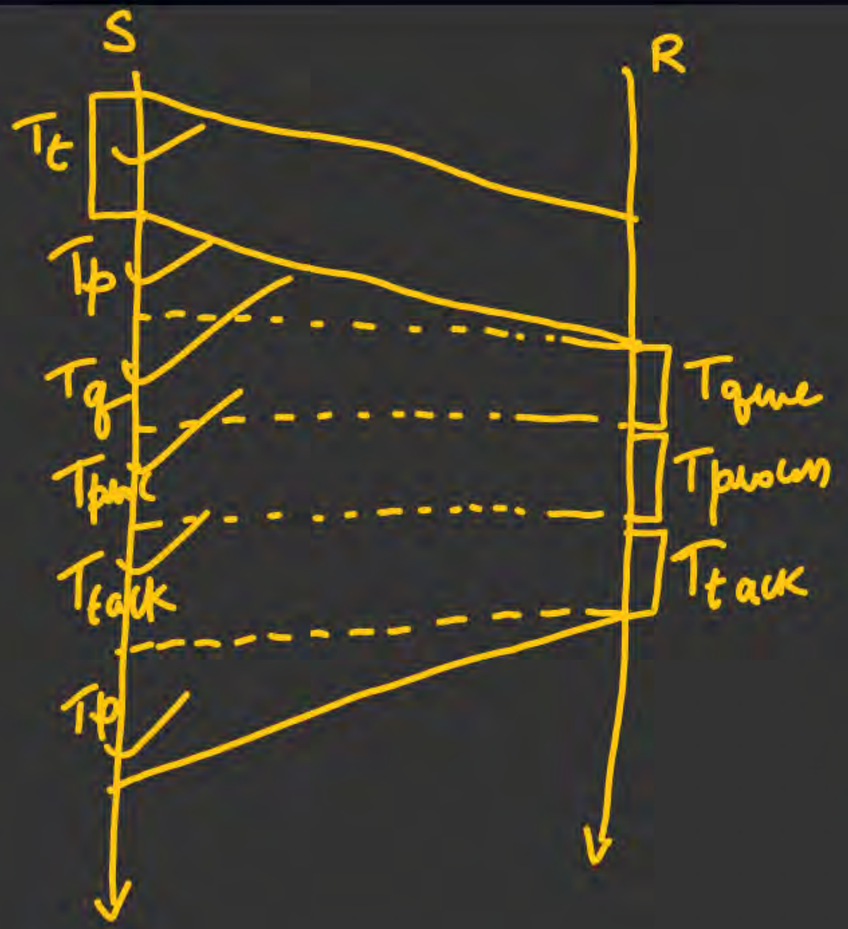


$$T_t + T_p + T_{queing} + T_{proc} + T_{tack} + T_p$$

Stop and wait:-

Sender will send a packet and waits for ack from the Rec before sending the new packet.





Total time = $T_t + T_p + T_q + T_{proc} + T_{t\,ack} + T_p$

T_t = L/B, T_p = d/v, $T_{t\,ack}$ = L_A/B

ACK are very small

Total time = $T_t + 2 * T_p$ → To send a packet in Stop and wait.



$$\text{efficiency } \eta = \frac{\text{useful time}}{\text{Total time}}$$

$$= \frac{T_t/T_t}{T_t/T_t + 2*T_p/T_t}$$

$$= \frac{1}{1 + 2\,(T_p/T_t)} \rightarrow a$$

$$\eta = \frac{1}{1+2a}$$

Throughput :- no of bits we are able to send in one sec

Packet Size $\leftarrow L_{bits} \longrightarrow T_t + 2 * T_p$ (sec)

1 sec $\longrightarrow \dfrac{L}{T_t + 2 * T_p}$

$= \eta * BW$

Throughput ✓

↑

Effective BW ✓ $= \eta * B$ $\dfrac{(L/B) * B}{T_t + 2 * T_p} = \dfrac{T_t'}{T_t + 2 * T_p}$ (B)

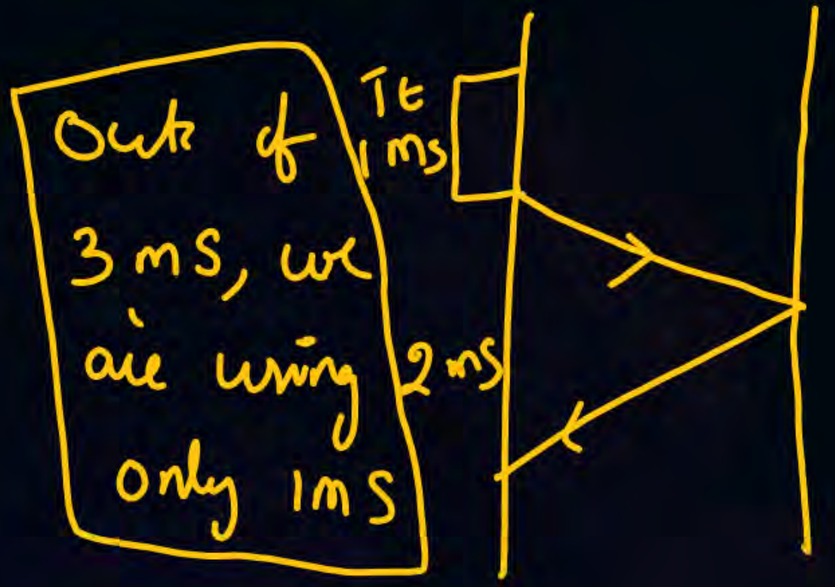
↑

BW utilization ✓

$= \dfrac{1}{1 + 2a}$ (B) $= \eta * BW$

$T_t = 1\,msec \quad T_p = 1\,msec$

$$\eta = \frac{1}{1+2a} = \frac{1}{1+2*\frac{T_p}{T_t}} = \frac{1}{1+2*1/1} = \frac{1}{3} = 33.33\,\%$$



Round trip time = $2 * T_p$

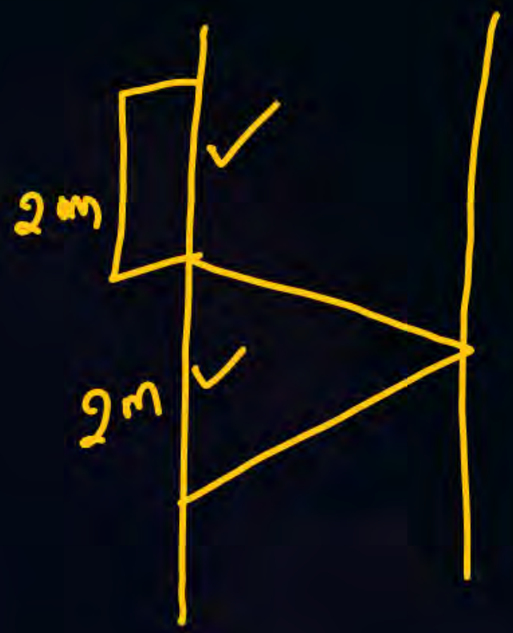
$$\frac{1\ ms}{3\ ms} = 33.33\%$$

Th if BW = 3 mbps

$Th = \eta \times BW = \frac{1}{3} \times 3\,mbps = 1\,mbps$

$T_t = 2ms \quad T_p = 1ms$

$\eta = ?$ $\quad \eta = \frac{1}{1+2a} = \frac{1}{1+2*T_p/T_t} = \frac{1}{1+2*1/2} = \frac{1}{2} = 50\%$



$BW = 4mbps$

Th put
or
eff BW
or
BW utilization $= \eta * BW = 1/2 \times 4mbps$

$= 2mbps$

⇒ If $\eta \geq 50\%$, then what is the relationship b/w T_p and T_t



$$\frac{T_t}{T_t + 2 * T_p} \geq \frac{1}{2}$$

$$2T_t \geq T_t + 2 * T_p$$

$$\Rightarrow \boxed{T_t \geq 2 * T_p}$$

$$\frac{L}{B} \geq 2 \times T_p$$

$$\boxed{L \geq 2 * T_p * B}$$

What is the packet size if η of stop and wait $\geq 50\%$

$$\eta = \frac{T_t}{T_t + 2 * T_p} \geq 1/2$$

Gate - CS

$$2T_t \geq T_t + 2 * T_p$$

$$T_t \geq 2 * T_p$$

$$\frac{L}{B} \geq 2 * T_p$$

$$L \geq 2 * T_p * BW$$

$\geq 50\%$

$BW = 4\,mbps \quad T_p = 1ms \quad L = ? \quad for\ \eta \geq 50\%$

mbps
10^6 bits

mBps
10^6 Byte

$L \geq 2 * T_p * BW$

$\geq 2 * 10^{-3}\,sec \times 4 \times 10^6\,bits/sec$

$\geq \underline{8000\,bits}$ ✓

$\neq 8K$ ✓

2^{10} ✓

$\eta = 1/2 \quad BW = 4\text{mbps}$

$$\text{Eff BW} = \eta \times BW$$
$$= 1/2 * 4\text{mbps}$$
$$= 2\text{mbps}$$

Factors effecting efficiency :-

$$\eta = \frac{1}{1+2a} = \frac{1}{1+2T_p/T_t} = \frac{1}{1+2(d/v)(B/L)}$$

$L \uparrow \quad \eta \uparrow$

Stop and wait is more efficient for bursty data

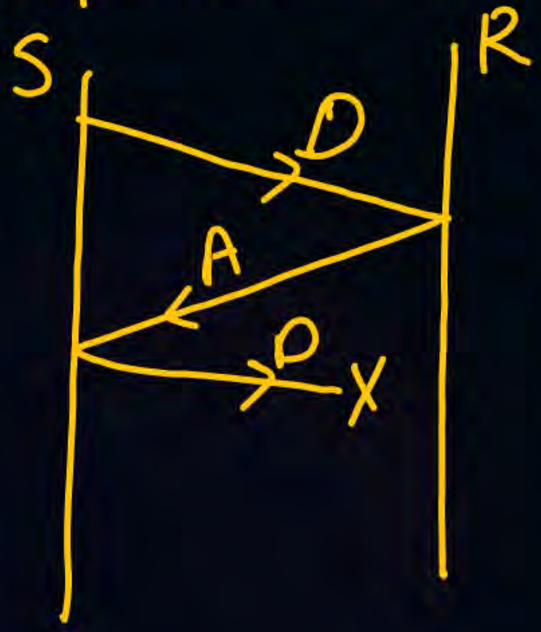
v – Constant
B – Constant
d – Variable
L – Variable

$d \uparrow \quad \eta \downarrow$

Stop and wait is useful in LANS.
Stop and wait is less efficient in WAN

Problems in Stop and wait:-

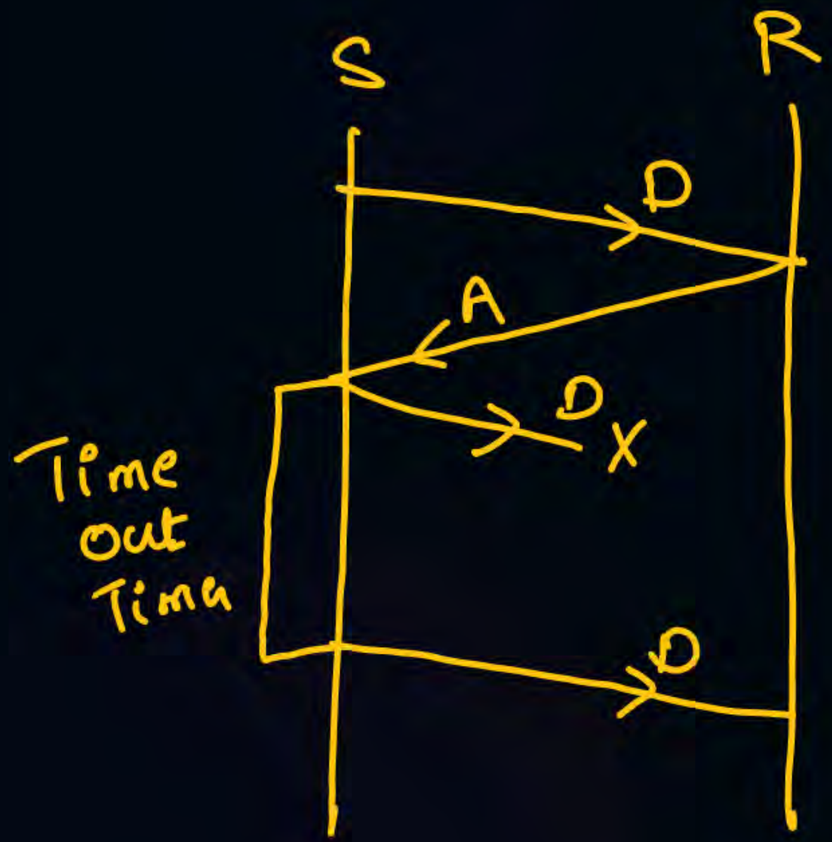
1) Data packet is lost:



Sender thinks that Rec is busy and so didn't Send ack

Rec thinks that Sender has not sent any data

Dead lock:-



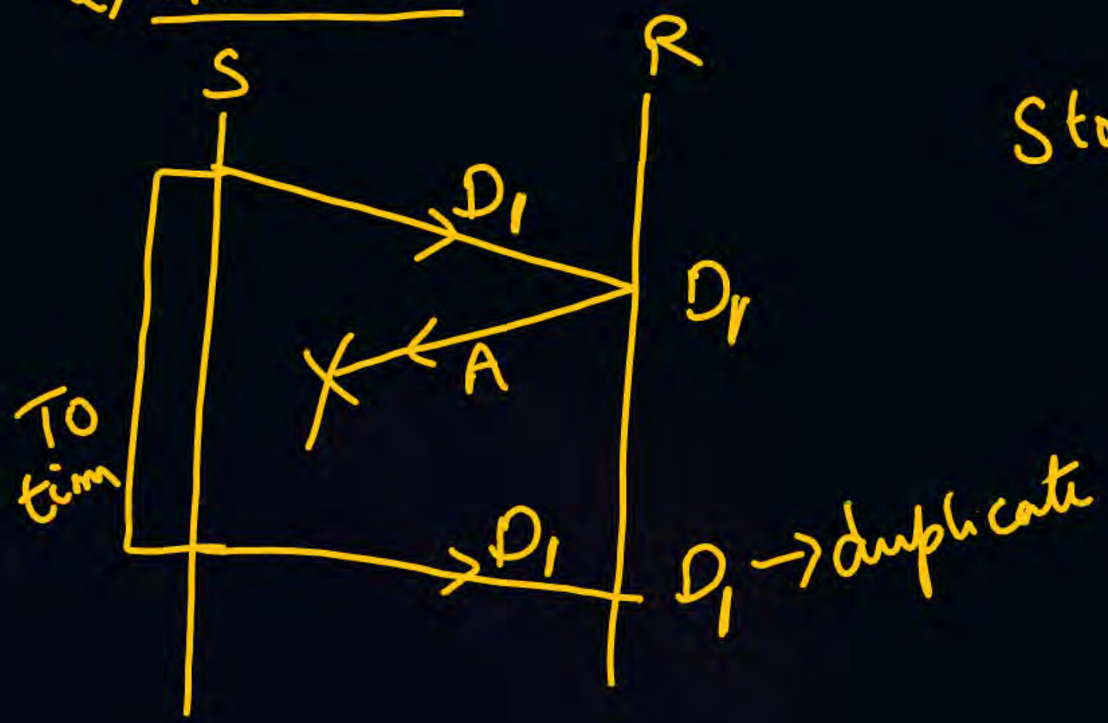
Data loss :-

Stop and wait + TO time

Stop and wait ARQ

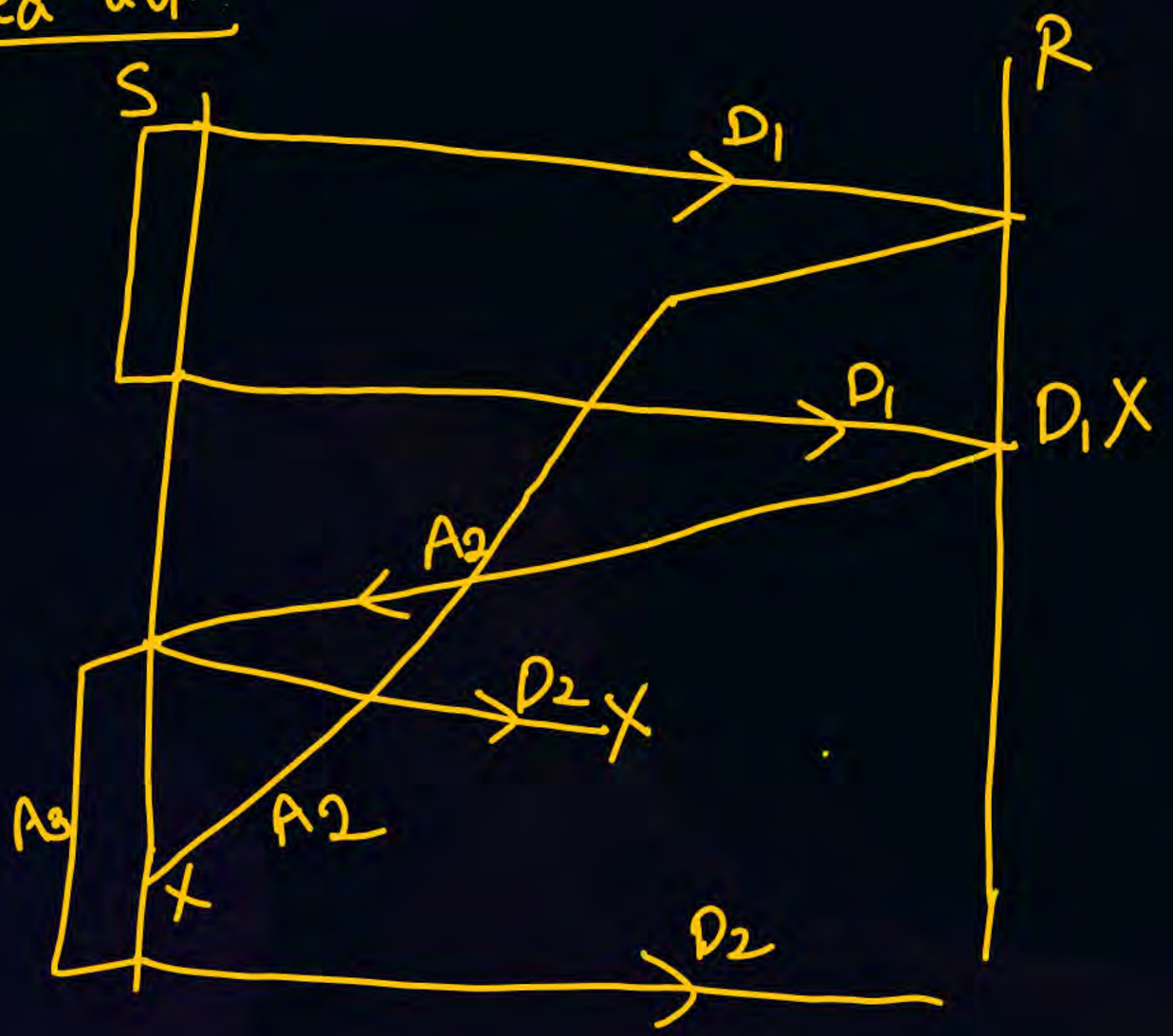
Automatic Repeat Request

2) Ack lost :-



Stop and wait ✓
+
TO time ✓
+
Data packets numbers ✓

3) Delayed ack :-



Stop and wait ✓
+
TO timer ✓
+
Data packets num ✓
+
Ack packets num ✓

→ we are sending 10 packets S→R using stop and wait and every 4th packet is lost, then how many packets are totally sent?

1 2 3 4 4 5 6 7 7 8 9 10 10 (13) ✓

Every 5th packet is lost

1 2 3 4 5 5 6 7 8 9 9 10

(12) ✓



error probability of the channel is 0.2.

If we send 400 packets, then how many packets are transmitted totally?

$$400 + 400(0.2) + 400(0.2)(0.2) + \cdots\cdots\cdots$$

$$n + np + np^2 + np^3 + \cdots\cdots$$

$$n(1 + p + p^2 + p^3 + \cdots\cdots) = \frac{n}{1-p} = \frac{400}{1-0.2} = \frac{400}{0.8} = 500$$

THANK - YOU